

```
In [1]: pip install faker
```

Requirement already satisfied: faker in c:\users\complab10\anaconda3\lib\site-packages (18.4.0)Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: python-dateutil>=2.4 in c:\users\complab10\anaconda3\lib\site-packages (from faker) (2.8.1)

Requirement already satisfied: six>=1.5 in c:\users\complab10\anaconda3\lib\site-packages (from python-dateutil>=2.4->faker) (1.15.0)

```
In [2]:
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from faker import Faker
```

```
In [3]:
```

```
fake= Faker()
Faker.seed(4321)
names_list=[]
for i in range(100):
    names_list.append(fake.name())
```

```
In [34]: names_list
```

```
Out[34]: ['Jason Brown',
          'Jacob Stein',
          'Cody Brown',
          'Larry Morales',
          'Jessica Hendricks',
          'Brian Moore',
          'Scott Baker',
          'Ruth Hoffman',
          'Daniel George',
          'David Moody',
          'Brian Maxwell',
          'Cory Cooper',
          'Morgan Foster',
          'James Sandoval',
          'Brian Lester',
          'Anne Davis',
          'Miss Amanda Harris MD',
          'Amy Dunn',
          'Brian Lee',
          ...]
```

```
In [5]:
```

```
np.random.seed(7)
salaries=[]
for i in range(100):
    salary=np.random.randint(1000,2500)
    salaries.append(salary)
```

```
In [6]: salaries
```

```
Out[6]: [1175,
          2220,
          1537,
          1502,
          1211,
          1919,
          2372,
          2209,
          2422,
          1535,
          1345,
          1366,
          1554,
          1730,
          1904,
          2191,
          2092,
          2456,
          1391,
          ...]
```

```
In [7]:
```

```
#Data Visualization
df=pd.DataFrame({'person':names_list,'salary':salaries})
df.head(3)
```

```
Out[7]:
```

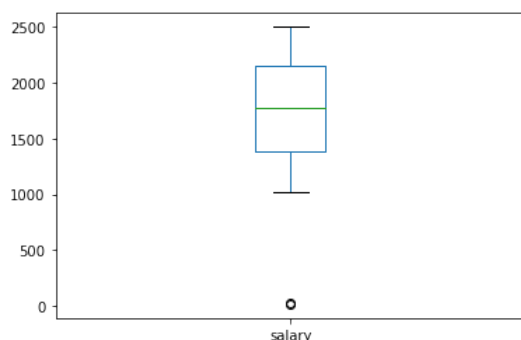
	person	salary
0	Jason Brown	1175
1	Jacob Stein	2220
2	Cody Brown	1537

```
In [8]: df.at[16,'salary']=23
df.at[65,'salary']=17
print(df.loc[16])
print(df.loc[65])
```

```
person    Miss Amanda Harris MD
salary                23
Name: 16, dtype: object
person      Joyce Bishop
salary                17
Name: 65, dtype: object
```

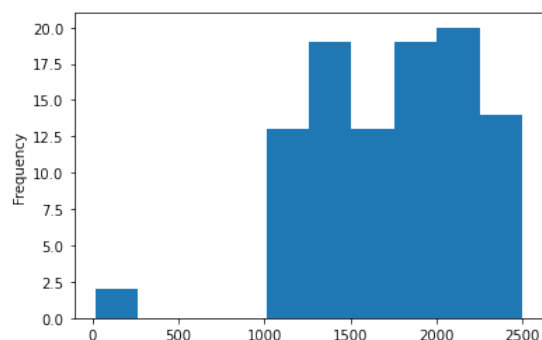
```
In [9]: df['salary'].plot(kind='box')
```

```
Out[9]: <AxesSubplot:>
```



```
In [10]: df['salary'].plot(kind='hist')
```

```
Out[10]: <AxesSubplot:ylabel='Frequency'>
```



```
In [11]: #unsupervised Learning
raw_salary=df['salary'].values
raw_salary
```

```
Out[11]: array([1175, 2220, 1537, 1502, 1211, 1919, 2372, 2209, 2422, 1535, 1345,
1366, 1554, 1730, 1904, 2191, 23, 2456, 1391, 1940, 2099, 1823,
1250, 2030, 2468, 2068, 2349, 2176, 1183, 1949, 2136, 1763, 2213,
1290, 1312, 1201, 2486, 1550, 1772, 1494, 2161, 2219, 2096, 1944,
1257, 1400, 2398, 2373, 1940, 1604, 1764, 1279, 1745, 1803, 2472,
2341, 2415, 2175, 1092, 1759, 2248, 1356, 1931, 1481, 1579, 17,
1481, 2043, 1923, 1787, 1033, 2147, 2310, 1741, 1989, 2439, 2154,
2107, 1402, 1021, 1203, 1047, 1784, 1524, 1349, 1107, 1393, 1844,
1622, 1654, 1636, 1276, 1309, 1827, 1035, 1260, 1456, 2498, 2007,
2239], dtype=int64)
```

```
In [12]: raw_salary=raw_salary.reshape(-1,1)
raw_salary
```

```
Out[12]: array([[1175],
 [2220],
 [1537],
 [1502],
 [1211],
 [1919],
 [2372],
 [2209],
 [2422],
 [1535],
 [1345],
 [1366],
 [1554],
 [1730],
 [1904],
 [2191],
 [ 23],
 [2456],
 [1391],
 ...])
```

```
In [13]: raw_salary=raw_salary.astype('float64')
raw_salary
```

```
Out[13]: array([[1175.],
 [2220.],
 [1537.],
 [1502.],
 [1211.],
 [1919.],
 [2372.],
 [2209.],
 [2422.],
 [1535.],
 [1345.],
 [1366.],
 [1554.],
 [1730.],
 [1904.],
 [2191.],
 [ 23.],
 [2456.],
 [1391.],
 ...])
```

```
In [14]: from sklearn.cluster import KMeans
cm=KMeans(n_clusters=4)
cm.fit(raw_salary)
```

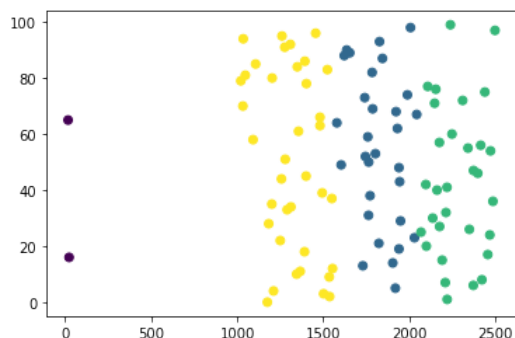
C:\Users\COMPLAB10\anaconda3\lib\site-packages\sklearn\cluster_kmeans.py:1416: FutureWarning: The default value of 'n_init' will change from 10 to 'auto' in 1.4. Set the value of 'n_init' explicitly to suppress the warning
super()._check_params_vs_input(X, default_n_init=10)
C:\Users\COMPLAB10\anaconda3\lib\site-packages\sklearn\cluster_kmeans.py:1440: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(

```
Out[14]: KMeans
KMeans(n_clusters=4)
```

```
In [15]: cm_labels=cm.labels_
```

```
In [16]: plt.scatter(raw_salary,np.arange(0,100),c=cm_labels)
```

```
Out[16]: <matplotlib.collections.PathCollection at 0x1d37e41fa30>
```



```
In [17]: #Supervised Learning
```

```
df['class']=0  
df['class']
```

```
Out[17]: 0      0  
         1      0  
         2      0  
         3      0  
         4      0  
         ..  
        95      0  
        96      0  
        97      0  
        98      0  
        99      0  
        Name: class, Length: 100, dtype: int64
```

```
In [19]: df.at[16,'class']=1  
df.at[65,'class']=1  
print(df.at[16,'class'])  
1
```

```
In [20]: df['class'].values
```

```
Out[20]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,  
                0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
                0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,  
                0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
                0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], dtype=int64)
```

```
In [21]: pip install pyod
```

Requirement already satisfied: pyod in c:\users\complab10\anaconda3\lib\site-packages (1.0.9)Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: numba>=0.51 in c:\users\complab10\anaconda3\lib\site-packages (from pyod) (0.53.1)

Requirement already satisfied: scikit-learn>=0.20.0 in c:\users\complab10\anaconda3\lib\site-packages (from pyod) (1.3.2)

Requirement already satisfied: numpy>=1.19 in c:\users\complab10\anaconda3\lib\site-packages (from pyod) (1.20.1)

Requirement already satisfied: matplotlib in c:\users\complab10\anaconda3\lib\site-packages (from pyod) (3.3.4)

Requirement already satisfied: scipy>=1.5.1 in c:\users\complab10\anaconda3\lib\site-packages (from pyod) (1.6.2)

Requirement already satisfied: six in c:\users\complab10\anaconda3\lib\site-packages (from pyod) (1.15.0)

Requirement already satisfied: joblib in c:\users\complab10\anaconda3\lib\site-packages (from pyod) (1.4.2)

Requirement already satisfied: llvmlite<0.37,>=0.36.0rc1 in c:\users\complab10\anaconda3\lib\site-packages (from numba>=0.51->pyod) (0.36.0)

Requirement already satisfied: setuptools in c:\users\complab10\anaconda3\lib\site-packages (from numba>=0.51->pyod) (52.0.0.post20210125)

Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\complab10\anaconda3\lib\site-packages (from scik

```
In [22]: from pyod.models.knn import KNN  
X=df['salary'].values.reshape(-1,1)  
y=df['class'].values  
clf=KNN(contamination=0.02,n_neighbors=5)  
clf.fit(X)
```

```
Out[22]: KNN(algorithm='auto', contamination=0.02, leaf_size=30, method='largest',  
            metric='minkowski', metric_params=None, n_jobs=1, n_neighbors=5, p=2,  
            radius=1.0)
```

```
In [23]: y_train_scores=clf.decision_scores_  
y_train_scores
```

```
Out[23]: array([[ 68.,  28.,  35.,  33.,  39.,  21.,  43.,  30.,  46.,  
                33.,  36.,  27.,  30.,  34.,  36.,  28., 1024.,  34.,  
                35.,  17.,  48.,  39.,  39.,  66.,  30.,  39.,  49.,  
                29.,  67.,  26.,  37.,  21.,  26.,  30.,  36.,  49.,  
                47.,  29.,  15.,  38.,  25.,  28.,  51.,  21.,  33.,  
                44.,  41.,  42.,  17.,  50.,  20.,  29.,  19.,  31.,  
                33.,  57.,  42.,  28.,  71.,  18.,  39.,  37.,  13.,  
                43.,  43., 1030.,  43.,  54.,  19.,  24.,  74.,  29.,  
                63.,  23.,  49.,  33.,  22.,  40.,  46.,  86.,  47.,  
                60.,  21.,  30.,  40.,  74.,  37.,  60.,  68.,  76.,  
                82.,  26.,  36.,  43.,  72.,  30.,  54.,  59.,  61.,  
                30.]])
```

```
In [25]: X_test=np.array([[35.]])  
clf.predict(X_test)
```

```
Out[25]: array([1])
```

```
In [32]: X_test2=np.array([[1005.]])  
clf.predict(X_test2)
```

```
Out[32]: array([0])
```

```
In [ ]:
```